

LECTURE 4

GROUP AND RING THEORY

4. MODULES - THE BASICS

4.1. Introduction. The concept of modules has wide-reaching applications, mainly within algebra (such as in Representation Theory) but also to other areas, such as category theory and geometry.

Definition 4.1. Let R be a ring. An (additive) abelian group M is a *left R -module* if there is a mapping $R \times M \rightarrow M$ given by $(r, m) \mapsto rm$, such that

- (i) $r(m_1 + m_2) = rm_1 + rm_2$;
- (ii) $(r_1 + r_2)m = r_1m + r_2m$;
- (iii) $(r_1r_2)m = r_1(r_2m)$;
- (iv) $1m = m$ if $1 \in R$,

for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$.

One sometimes denotes a left R -module M as ${}_R M$.

Example 4.2. If $R = F$ is a field, then M is just a vector space over F .

Often rm is just called the multiplication of m by r on the left. We define right R -modules similarly, via the mapping $M \times R \rightarrow M$, $(m, r) \mapsto mr$. If R is a commutative ring and M is a left R -module, then M can be made into a right R -module by defining $mr = rm$. In this case, abstractly speaking, a left R -module is the same as a right R -module. Thus, when R is commutative, we do not distinguish between left and right R -modules and simply call them R -modules.

Throughout, all modules are left modules unless otherwise stated.

Example 4.3. Let A be an abelian group. Then A is a left (also right) $\mathbb{Z}$ -module, because

$$\begin{aligned}k(a_1 + a_2) &= ka_1 + ka_2, \\(k_1 + k_2)a &= k_1a + k_2a, \\(k_1k_2)a &= k_1(k_2a), \\1a &= a,\end{aligned}$$

for all integers $k, k_1, k_2 \in \mathbb{Z}$ and for all $a, a_1, a_2 \in A$. [Recall that $ka = a + \cdots + a$ for $k \in \mathbb{N}$ and $a \in A$. If $k < 0$, then $ka = (-k)(-a)$.]

Example 4.4. Let R be a ring. Then R itself can be regarded as a left R -module by defining am , for $a, m \in R$, to be the product of a and m as elements of the ring R .

Then the distributive laws and the associative law for multiplication in the ring R show that R is a left R -module. Similarly R is also a right R -module.

Example 4.5. Let $M = M_{m \times n}(R)$ be the set of $m \times n$ matrices over a ring R . Then M is a module over R , because M is an abelian group under the usual addition of matrices, and

the usual scalar multiplication (ra_{ij}) of the matrix $(a_{ij}) \in M$ by the element $r \in R$ satisfies axioms (i) - (iv) for a module.

In particular, the set of $1 \times n$ (or $m \times 1$) matrices - the set of n -tuples denoted by R^n - is a module over R . Further by choosing $R = \mathbb{R}$ and $n = 2$ (or 3), we obtain that the set of vectors in a plane (or in space) forms a vector space over the field $\mathbb{R}$.

Example 4.6. (Direct product of modules) Let M and N be R -modules. In the Cartesian product $M \times N$, define

$$(x, y) + (x', y') = (x + x', y + y'),$$

$$r(x, y) = (rx, ry)$$

for all $x, x' \in M$, $y, y' \in N$, and $r \in R$. Then $M \times N$ is an R -module, called the direct product of the R -modules M and N . [Sometimes the direct product is written $M \oplus N$.]

We end this section with some elementary properties of an R -module.

Lemma 4.7. *Let M be an R -module. Then*

- (i) $0m = 0$, for $m \in M$;
- (ii) $a0 = 0$, for $a \in R$;
- (iii) $(-a)m = -(am) = a(-m)$, for $a \in R$ and $m \in M$,

where 0 on the right sides of (i) and (ii) is the zero of M , and 0 on the left side of (i) is the zero of R .

Proof. To prove (i), consider $am = (a + 0)m = am + 0m$. To prove (ii), consider $am = a(m + 0) = am + a0$. To prove (iii), consider $0 = 0m - (a + (-a))m = am + (-a)m$, and also consider $0 = a0 = a(m + (-m)) = am + a(-m)$. $\square$

4.2. Submodules and direct sums.

Definition 4.8. A subset N of an R -module M is called an R -submodule (or simply submodule) of M if

- (i) $N \neq \emptyset$;
- (ii) $a - b \in N$ for all $a, b \in N$;
- (iii) $ra \in N$ for all $a \in N$, $r \in R$.

Clearly (0) , or simply 0 , and M are R submodules, called *trivial submodules*. In the case when R is a field, the subset N is called a *subspace* of M .

We remark that if N is an R -submodule of M , then N is also an R -module in its own right.

Example 4.9. Each left ideal of a ring R is an R -submodule of the left R -module R , and conversely. This follows from the definition of a left ideal and from the ring axioms.

Certainly, also all subspaces of a vector space yield examples of submodules.

Example 4.10. If M is an R -module and $x \in M$, then the set

$$Rx = \{rx \mid r \in R\}$$

is an R -submodule of M , for

$$r_1x - r_2x = (r_1 - r_2)x \in Rx,$$

$$r_1(r_2x) = (r_1r_2)x \in Rx,$$

for all $r_1, r_2 \in R$. Clearly $Rx \neq \emptyset$, as $0 \in R$ and so $0x = 0 \in Rx$.

Example 4.11. If M is an R -module and $x \in M$, then the set

$$K = \{rx + nx \mid r \in R, n \in \mathbb{Z}\}$$

is an R -submodule of M containing x . Further, if R has a unity, then $K = Rx$.

Solution: First, we observe that $(K, +)$ is an abelian subgroup of $(M, +)$. Next, let $a \in R$ and consider $rx + nx \in K$. Then

$$\begin{aligned} a(rx + nx) &= a(rx) + a(nx) \\ &= \begin{cases} (ar)x + a(x + \cdots + x) & \text{if } n > 0 \\ (ar)x + a((-x) + \cdots + (-x)) & \text{if } n < 0, \end{cases} \end{aligned}$$

and

$$a(rx + nx) = \begin{cases} ((ar) + a + \cdots + a)x & \text{if } n > 0 \\ ((ar) + (-a) + \cdots + (-a))x & \text{if } n < 0, \end{cases}$$

since $a(-x) = (-a)x$. Therefore,

$$a(rx + nx) = ux \quad \text{for some } u \in R.$$

Hence $a(rx + nx) \in K$, for all $a, r \in R$ and for all n (including 0) in $\mathbb{Z}$. Choosing $r = 0 \in R$ and $n = 1 \in \mathbb{Z}$ in $rx + nx$ gives $x \in K$. It is worth noting that if L is any other R -submodule of M containing x , then L contains all elements of the form $rx + nx$, for $r \in R, n \in \mathbb{Z}$. Hence $K \subseteq L$. Thus K is the smallest R -submodule of M containing x , usually denoted (x) .

Suppose now that R has the unity 1. Trivially $Rx \subseteq K$, so it remains to show that $K \subseteq Rx$. Let $rx + nx \in K$. For $n > 0$,

$$\begin{aligned} rx + nx &= rx + n(1x) = rx + (1x + \cdots + 1x) \\ &= (r + 1 + \cdots + 1)x = ux \end{aligned}$$

for some $u \in R$. Thus $rx + nx \in Rx$. Similarly, if $n \leq 0$, then $rx + nx \in Rx$. Hence the result.

Theorem 4.12. Let $(N_i)_{i \in I}$ be a family of R -submodules of an R -module M . Then $\cap_{i \in I} N_i$ is also an R -submodule.

Proof. We have $0 \in \cap_{i \in I} N_i$, so $\cap_{i \in I} N_i \neq \emptyset$. Let $x, y \in \cap_{i \in I} N_i$, and $r \in R$. Then for all $i \in I$, we have $x - y \in N_i$ and $rx \in N_i$, since the N_i are R -submodules. Thus $x - y$ and rx are in $\cap_{i \in I} N_i$, which proves that $\cap_{i \in I} N_i$ is an R -submodule. $\square$

Let S be a subset of an R -module M , and let

$$\mathcal{A} = \{N \mid N \text{ is an } R\text{-submodule of } M \text{ containing } S\}.$$

Then $\mathcal{A} \neq \emptyset$ because $M \in \mathcal{A}$. Let $K = \cap_{N \in \mathcal{A}} N$. Then K is the smallest R -submodule of M containing S and is denoted by (S) .

Definition 4.13. The smallest R -submodule of M containing a subset S is called the R -submodule *generated by* S . If $S = \{x_1, \dots, x_m\}$ is a finite set, then (S) is also written as $(x_1, \dots, x_m)$.

An R -module M is called a *cyclic module* if $M = (x)$ for some $x \in M$.

Example 4.14. Finitely generated modules over $\mathbb{Z}$ coincide with the finitely generated abelian groups.

An example of a cyclic module is $2\mathbb{Z}$ as a $\mathbb{Z}$ -module. Also, every cyclic group is a cyclic $\mathbb{Z}$ -module.

One can check that a cyclic module generated by x is precisely $\{rx + nx \mid r \in R, n \in \mathbb{Z}\}$, and if R is with unity then it simplifies to $\{rx \mid r \in R\}$, that is, to Rx .

More generally, for an R -module M and $x_1, \dots, x_n \in M$, we write

$$\sum_{i=1}^n Rx_i := \{r_1x_1 + r_2x_2 + \cdots + r_nx_n \mid r_i \in R\}.$$

Theorem 4.15. *Let R be a ring with unity. We have $(x_1, \dots, x_n) = \sum_{i=1}^n Rx_i$.*

Proof. It is clear that $(x_1, \dots, x_n) \supseteq \sum_{i=1}^n Rx_i$, since $(x_1, \dots, x_n)$ is an R -module and $x_1, \dots, x_n \in (x_1, \dots, x_n)$. For the reverse containment, it suffices to show that $\sum_{i=1}^n Rx_i$ is a submodule, since by definition $(x_1, \dots, x_n)$ is the smallest submodule containing $x_1, \dots, x_n$.

Clearly $0 = \sum_{i=1}^n 0x_i \in \sum_{i=1}^n Rx_i$, so $\sum_{i=1}^n Rx_i \neq \emptyset$.

Let $\sum_{i=1}^n r_ix_i, \sum_{i=1}^n s_ix_i \in \sum_{i=1}^n Rx_i$ and let $r \in R$. Then

$$\sum_{i=1}^n r_ix_i - \sum_{i=1}^n s_ix_i = \sum_{i=1}^n (r_i - s_i)x_i \in \sum_{i=1}^n Rx_i,$$

and

$$r\left(\sum_{i=1}^n r_ix_i\right) = \sum_{i=1}^n r(r_ix_i) = \sum_{i=1}^n (rr_i)x_i \in \sum_{i=1}^n Rx_i,$$

so $\sum_{i=1}^n Rx_i$ is an R -submodule. □

We now proceed to define some concepts needed to talk about direct sums of modules.

Definition 4.16. Let M be an R -module. If an element $m \in M$ can be expressed as $m = a_1x_1 + \cdots + a_nx_n$, with $a_i \in R$ and $x_i \in M$, for $i \in \{1, \dots, n\}$, then we say that m is a *linear combination* of the elements $x_1, \dots, x_n$ over R .

Remark 4.17. The set of generators of a module need not be unique, as seen in Linear Algebra that a basis of a vector space is not unique. Similarly, we know that the generators of an abelian group are not unique, e.g. $\mathbb{Z} = \langle 1 \rangle = \langle -1 \rangle$.

Definition 4.18. Let (N_i) , for $1 \leq i \leq k$, be a family of R -submodules of a module M . Then the submodule generated by $\bigcup_{i=1}^k N_i$, that is, the smallest submodule containing the submodules N_i is called the *sum of the submodules N_i* , and is denoted by $\sum_{i=1}^k N_i$.